

Question 1:

Source code:

```
#include <iostream>
using namespace std;
class BankAccount{
    private:
        static int totalbalance;
    public:
        int depositMoney;

        BankAccount(int d){
            depositMoney = d;
            totalbalance+=depositMoney;
            cout<<"AMOUNT ADDED: "<<depositMoney<<endl;
        }

        ~BankAccount(){
            cout<<"ACCOUNT CLOSED!"<<endl;
        }

        static void display_balance(){
            cout<<"TOTAL BALANCE: "<<totalbalance<<endl;
        }
};
```

```
int BankAccount::totalbalance=0;
int main(){
    BankAccount::display_balance();
    BankAccount BA1(200);
    BA1.display_balance();

    BankAccount BA2(2000);
    BA2.display_balance();
    return 0;
}
```

Output:

```
TOTAL BALANCE: 0
AMOUNT ADDED: 200
TOTAL BALANCE: 200
AMOUNT ADDED: 2000
TOTAL BALANCE: 2200
ACCOUNT CLOSED!
ACCOUNT CLOSED!
```

Question 2:

Source code:

```
#include <iostream>
using namespace std;
class libraryBook{
private:
    static int bookCounter;
public:
    libraryBook(){
        if(bookCounter<5){
            bookCounter++;
        }
        else{
            cout<<"LIBRARY IS FULL, CAN'T ADD MORE BOOKS!"<<endl;
        }
    }

    ~libraryBook(){
        bookCounter--;
    }

    static void display_bookCount(){
        cout<<"BOOK COUNT: "<<bookCounter<<endl;
    }
};
```

```

int libraryBook::bookCounter=0;

int main(){
    libraryBook b1,b2,b3,b4;

    b4.display_bookCount();
    libraryBook b5;
    b5.display_bookCount();
    libraryBook b6;
    b6.display_bookCount();

    return 0;
}

```

Output:

```

BOOK COUNT: 4
BOOK COUNT: 5
LIBRARY IS FULL, CAN'T ADD MORE BOOKS!
BOOK COUNT: 5

```

Question 3:

Source code:

```

#include <iostream>
using namespace std;
class Employee{
private:
    static int empCount;
public:
    Employee();
    ~Employee();
    static void display_empCount();
};

```

```

#include<iostream>
#include "class_header.h"
Employee::Employee(){
    if(empCount<10){
        empCount++;
    }
    else{
        cout<<"NO MORE EMPLOYEES CAN BE ADDED!"
    }
}
Employee::~~Employee(){
    empCount--;
}
void Employee::display_empCount(){
    cout<<"EMPLOYEE COUNT: "<<empCount<<endl;
}
int Employee::empCount=0;

```

```

#include <iostream>
#include "class_header.h"
using namespace std;

int main(){
    Employee emp1,emp2,emp3,emp4,emp5,emp6,emp7,emp8,emp9;
    Employee::display_empCount();
    Employee emp10;
    Employee::display_empCount();
    Employee emp11;
    Employee::display_empCount();
    return 0;
}

```

Output:

```
EMPLOYEE COUNT: 9  
EMPLOYEE COUNT: 10  
NO MORE EMPLOYEES CAN BE ADDED!  
EMPLOYEE COUNT: 10
```